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InstantHMR: Real-Time Human Mesh Recovery on Your Phone

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Abstract

Real-time 3D human mesh recovery from a single image on mobile devices is a challenging problem with many applications, including robotics, biomechanics, AR/VR, and exergames. Current state-of-the-art models are either too slow to run efficiently on mobile devices or sacrifice accuracy for speed. We present InstantHMR, a family of open-source single-image regressors suited for mobile inference, based on the Momentum Human Rig (MHR) parametric model and trained exclusively on publicly available annotations from the SAM 3D Body dataset. Our approach achieves remarkable performance on a Samsung Galaxy S23 NPU, recovering a 6,890-vertex body mesh in 5.5 ms per person, and 0.98 ms on an NVIDIA RTX 4070 GPU. This speed surpasses all existing methods we evaluated. On mobile, it is 12× and 23× faster than NLF-S and NLF-L, the only phone-compatible methods that outperform ours in accuracy, while achieving competitive accuracy with 40.0 mm PA-MPJPE on 3DPW, 48.8 mm on EMDB and 41.7 mm on RICH. To ensure fair comparisons, we re-evaluate 11 state-of-the-art methods under a unified protocol aligned with the closest MHR-based competitors, including Meta's official MHR-to-SMPL conversion, and report their reproduction results. Additionally, we measure latency across eight different targets, ranging from desktop TensorRT to two phone NPUs. Code and models are available at https://github.com/Naturalpad-studio/InstantHMR.

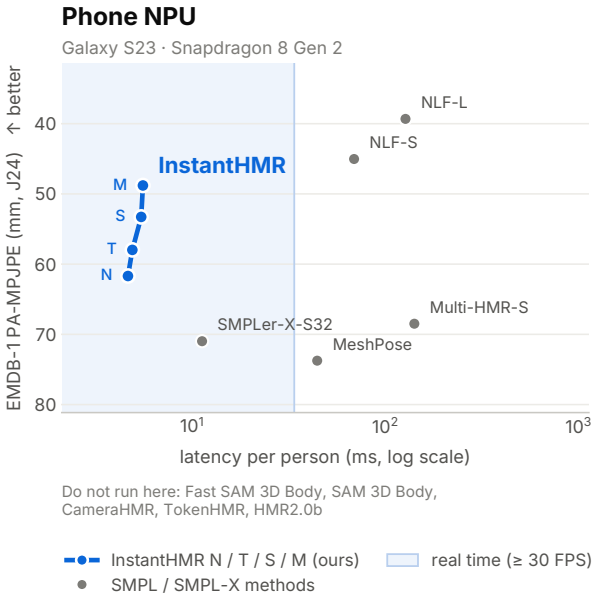

Figure 1. **Accuracy against latency on a phone.** EMDB-1 PA-MPJPE (24 joints, lower is better, up) against model latency per person on the NPU of a Samsung Galaxy S23 (log scale, faster is left), for every method under the protocol of Sec. 4. InstantHMR's four sizes form the blue curve. The shaded zone is real time (≥ 30 FPS): the only other method in it, SMPLer-X, is 22 mm less accurate than InstantHMR-M, and the methods more accurate than InstantHMR-M run 12–23× slower or not at all. The same plot on a desktop GPU is in the supplementary material.

1. Introduction

Monocular real-time human mesh recovery (HMR) is a long-standing goal in computer vision, with applications in robotics [10], tele-rehabilitation [39], workplace ergonomics [13, 21], AR/VR [12, 27], exergames [45], and 3D animation [37]. Since 2023, most smartphones include a neural processing unit (NPU), which is a formidable opportunity for local real-time inference, fostering privacy, low energy consumption and ease of use.

Recent models recover a body mesh from a single image with remarkable robustness [29, 34, 44], but they are built for GPUs: SAM 3D Body [44] takes 416 ms per person on a desktop GPU in its own release and does not run on a phone's NPU at all. NLF [34] is the fastest state-of-the-art model on a desktop GPU (1.52 ms per person for the smallest variant), but is not tailored for real-time mobile inference (67.97 ms per person on a Samsung Galaxy S23 NPU). On the other hand, models like SMPLer-X-S32 or MeshPose [4, 18] that run real-time[1] on a phone give up much of the

[1]In this context, real-time means a latency lower than 33 ms, 30 fps

accuracy (cf. Tab. 2 and Fig. 1).

We present InstantHMR, a family of single-image HMR models designed for phones. InstantHMR regresses the parameters of the Momentum Human Rig (MHR) [7] with an efficient convolutional backbone [46] and a small transformer decoder. It is trained on the publicly released annotations of SAM 3D Body's training set. On a Samsung Galaxy S23 NPU, InstantHMR-M recovers a 6,890-vertex mesh in 5.5 ms per person, faster than every method we measure and 12× and 23× faster than NLF-S and NLF-L, the only phone-compatible methods that outperform ours in accuracy (cf. Fig. 1 and Tab. 1). Published numbers on EMDB [15] and 3DPW [40] are produced under different person boxes, focal lengths, joint sets and conversions between body models, and these choices move a single model by several millimetres (supplementary material), as much as the gaps between methods. We therefore re-evaluate eleven methods under one protocol, matched to the closest MHR-based competitors, and report for each how closely we reproduce its paper. In summary, our contributions are:

- **InstantHMR**, a family of four MHR regressors (nano to medium) that trace one accuracy/latency curve, trained only on released annotations; InstantHMR-M runs in real time on a phone NPU at 48.8 mm PA-MPJPE on EMDB-1, 40.0 mm on 3DPW and 41.7 mm on RICH.
- **A matched evaluation protocol** with Meta's official MHR-to-SMPL conversion, under which eleven competitors are re-scored, each with a reproduction verdict against its own paper.
- **Latency on eight targets**, from desktop TensorRT to two phone NPUs and two phone CPUs, measured the same way for every method.

The code (Apache-2.0) and the models (for non-commercial use) are available at https://github.com/Naturalpad-studio/InstantHMR.

We also provide a commercially safe variant of the medium model (InstantHMR-M-C) trained on a smaller corpus of images with a permissive license (cf. Tab. 3).

2. Related work

Parametric body models. Most HMR methods do not predict a mesh directly, but the parameters of a parametric body model that maps a few pose and shape coefficients to a skinned mesh. SMPL [25] is by far the most used, with a 6,890-vertex mesh, 24 joints and a shape space learned from 3D scans. SMPL-H [32] adds articulated hands, and SMPL-X [30] adds both hands and an expressive face. More recently, Meta released the Momentum Human Rig (MHR) [7], which decouples the skeleton from the body shape. Unlike SMPL, Anny [3] does not learn its shape space from 3D scans. Its shape is set by interpretable parameters such as age, height and weight, built on the open-source MakeHuman software and calibrated on WHO population statistics, so that a single model spans bodies from infants to elders. SOMA [33] goes one step further and maps SMPL, SMPL-X, MHR and Anny to a single canonical mesh and skeleton. These models differ in topology, skeleton and units, so comparing methods that predict different models requires converting between them, which is a key part of our protocol (Sec. 4). InstantHMR predicts in MHR parametric space.

Human mesh recovery from a single image. Historically, HMR [14] consists in regressing SMPL parameters from a single image. HMR2.0 [8] scales this formulation with a ViT backbone and large pseudo-labelled data, TokenHMR [6] predicts the pose as discrete tokens, CLIFF [23] conditions the regressor on the location of the person in the full frame, and CameraHMR [29] adds a camera-aware head trained with estimated focal lengths. HybrIK [19] recovers the rotations from 3D keypoints with analytical inverse kinematics, and NLF [34] predicts any body point as a continuous field, which allows training a single model on annotations from different skeletons and meshes. SMPLer-X [4] and Multi-HMR [2] regress the expressive SMPL-X model, for one person or for many at once, and the authors of Anny show that a Multi-HMR model trained on synthetic Anny images matches its scan-based counterparts [3]. SAM 3D Body [44] moves to MHR with a promptable DINOv3-based architecture trained on a large human-annotated corpus, and Fast SAM 3D Body [43] accelerates it by up to 10.9× without retraining. NVIDIA's GEM-X [28] also builds on DINOv3 and predicts SOMA. It provides a single-image model, and a video model that we discuss in the next paragraph. These recent models are remarkably robust, but they are built for GPUs. InstantHMR predicts in MHR parametric space, with an architecture tailored to phone inference.

Human mesh recovery from video. Video methods use the temporal information to recover smoother and more accurate motion, often in world coordinates. WHAM [38] lifts sequences of 2D keypoints to 3D, fuses them with the features of a pretrained single-image encoder, and estimates the global trajectory with recurrent networks that only look at past frames, so it runs online. TRAM [41] and GVHMR [36] recover the global trajectory and motion of people from whole sequences, GEM [20] unifies motion estimation and generation in a single diffusion-based model, and the video model of GEM-X [28] follows the motion over the sequence from the features of its single-image model and 2D keypoints. FastHMR [26] accelerates a ViT regressor by merging tokens and layers, then denoises the pose over 243-frame clips with a diffusion decoder, reaching 103–150 FPS on an RTX 3090.

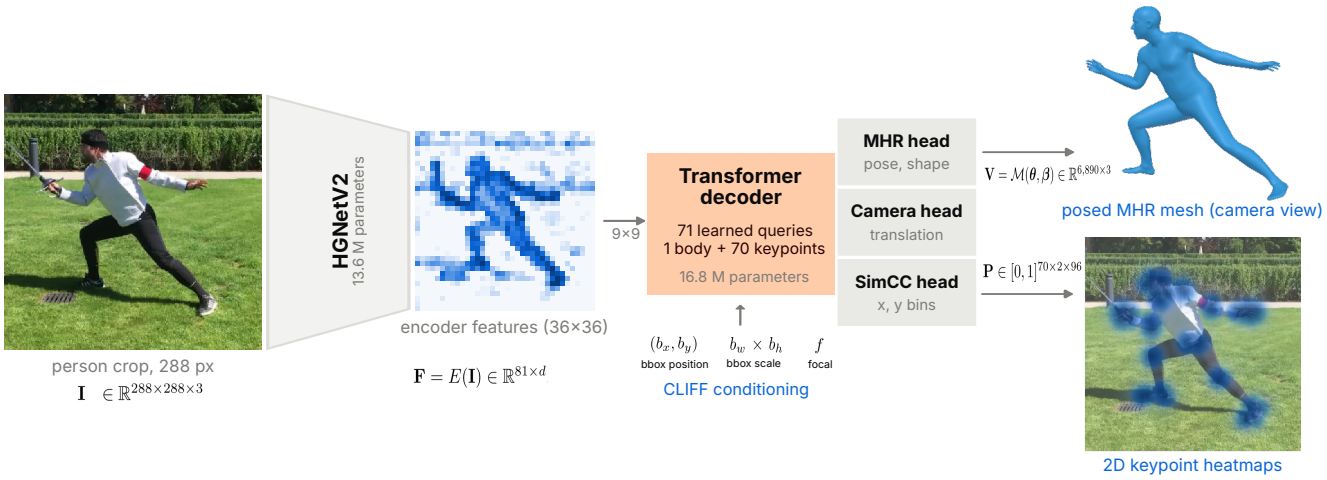

Figure 2. **InstantHMR.** A 288 px crop around the person goes through a convolutional HGNetV2 encoder (shown: the first principal component of its second-stage features, 36×36, contrast-stretched) down to 9×9 image tokens. A 4-layer transformer decoder refines 71 learned queries (one for the body, one per keypoint) by attention among themselves and to the image tokens; the box's position in the full frame (two viewing angles and the size over the focal length) is embedded and added to every query. The body query feeds the MHR and camera heads, from which the rig's forward kinematics (FK) gives the posed mesh; each keypoint query feeds a SimCC head that classifies its x and y over 96 bins (shown over the crop: for each keypoint, the product of its two distributions). Every visual is the real output of InstantHMR-M on a 3DPW test frame [40]. InstantHMR-M has 32.2 M parameters: 13.6 M in the backbone, 16.8 M in the decoder, and 1.8 M in the projections, queries and heads.

Efficient and on-device recovery. Many applications such as AR/VR, tele-rehabilitation or the teleoperation of robots, need a low latency and not only a high throughput: the pose of a frame must be available while that frame is captured. Most video methods induce latency, since an output that depends on future frames has to wait for them: FastHMR waits for up to 242 later frames, about 8 s at 30 FPS. Online methods such as WHAM avoid this wait, but still need a per-frame image model: the 200 FPS reported for WHAM exclude its ViTPose keypoint detector and its pretrained image encoder [38]. This study therefore focuses on the single-image model, which estimates the current frame from that frame alone and is the per-frame building block of online video methods such as WHAM. Among single-image methods, efficient HMR has mostly targeted GPUs, *e.g.* Seeing Machines reports 5.5–15.8 ms on an NVIDIA Jetson Thor robotics module [35]. On phones, BlazePose GHUM Holistic [9] estimates 3D landmarks in real time and lifts them to a body model, but is not evaluated on a standard mesh benchmark. The closest work to ours is MeshPose [18], which localises a low-poly SMPL mesh with DensePose-style supervision and runs at 97–153 FPS on an iPhone 12, depending on the backbone; we evaluate its released model in Sec. 5.

Evaluation protocols. 3DPW [40], EMDB [15] and RICH [11] are the standard in-the-wild benchmarks, but papers evaluate on them in different ways. They differ in the person boxes (ground truth or detections), the focal length (calibrated or a fixed field of view), and, for methods that do not output SMPL, the conversion to SMPL. For instance, SAM 3D Body converts MHR to SMPL with Meta's iterative fitter, Fast SAM 3D Body replaces that fitter with a learned network, and GEM-X releases its own dataset-specific projectors. Each of these choices moves a single model by several millimetres, so we fix one protocol, the one of the closest MHR-based competitors, and re-run every method under it (Sec. 4).

3. InstantHMR

InstantHMR takes a square crop of the person, $\mathbf{I} \in \mathbb{R}^{288\times288\times3}$, and the position of that crop in the full image, $\mathbf{c} \in \mathbb{R}^3$ (defined in Sec. 3.1). It returns three outputs (Fig. 2). The first is the person's MHR parameters: the rig parameters $\boldsymbol{\theta} \in \mathbb{R}^{204}$, which hold the root translation and rotation, 130 joint parameters and 68 bone scales, and the identity coefficients $\boldsymbol{\beta} \in \mathbb{R}^{45}$, which shape the body surface. The second is, for each of the 70 2D keypoints, two probability distributions over 96 bins, one for its x and one for its y coordinate, $\mathbf{P} \in [0,1]^{70\times2\times96}$ (SimCC [22]). Their expectations give the 2D keypoints $\mathbf{x} \in \mathbb{R}^{70\times2}$, in crop coordinates. The third is the camera translation $\mathbf{t} \in \mathbb{R}^3$, which places the body in the camera frame. The MHR rig $\mathcal{M}$ then turns $(\boldsymbol{\theta}, \boldsymbol{\beta})$ into the posed mesh $\mathbf{V} = \mathcal{M}(\boldsymbol{\theta}, \boldsymbol{\beta}) \in \mathbb{R}^{6,890\times3}$ and its 70 3D keypoints $\mathbf{J} \in \mathbb{R}^{70\times3}$, which are in camera coordinates once translated by $\mathbf{t}$. $\mathbf{V}$ is a farthest-point sub-

set of MHR's 18,439-vertex mesh, with as many vertices as SMPL, so that every method's latency is measured up to a mesh of the same size. Every design choice serves one constraint: the whole network must compile to a phone NPU, in fp16, with no operator falling back to the CPU. The mathematical details of this section, the equations of the network and of the losses, are given in the supplementary material.

3.1. Network

HGNetV2 Backbone. A convolutional HGNetV2 [46] backbone E, pretrained on ImageNet, turns the 288×288 crop into a 9×9 grid of features. A linear layer projects them to $d = 512$ channels, and a fixed 2D sine-cosine positional embedding is added, which gives the 81 image tokens $\mathbf{F} = E(\mathbf{I}) \in \mathbb{R}^{81 \times d}$. Convolutions, unlike the attention of a ViT backbone, map directly onto the operators phone NPUs accelerate. The four sizes of InstantHMR differ only in the backbone (B0, B1, B2, B4 for nano, tiny, small, medium), and have 20.0, 20.3, 24.4 and 32.2 M parameters. The decoder and the heads are the same in every size, and the decoder alone holds 16.8 M of these parameters.

CLIFF conditioning. A crop alone does not determine where the person is in 3D: the same crop could come from the centre or the edge of a wide-angle image. As in CLIFF [23], the network therefore receives the position of the crop in the full image, but expressed through the focal length f rather than the image size, so that it keeps one meaning across cameras. For a box of centre (b_x, b_y) and size $b_w \times b_h$ in an image of size $W \times H$, $\mathbf{c}$ holds the two viewing angles of the box centre, $\arctan((b_x - W/2)/f)$ and $\arctan((b_y - H/2)/f)$, and the box size $\max(b_w, b_h)/f$. A two-layer MLP ϕ embeds it into $\phi(\mathbf{c}) \in \mathbb{R}^d$. At deployment, without calibration, f defaults to $1.05\times$ the image diagonal.

Transformer Decoder. 71 learned queries $\mathbf{Q} \in \mathbb{R}^{71 \times d}$, one for the body and one per 2D keypoint, receive the position embedding $\phi(\mathbf{c})$, added to every query. A 4-layer pre-norm transformer decoder [5] D (8 heads) then lets them attend to each other and to the image tokens $\mathbf{F}$, and outputs one feature $\mathbf{z}_i \in \mathbb{R}^d$ per query. The feature $\mathbf{z}_0$ comes from the body query, and $\mathbf{z}_1, \ldots, \mathbf{z}_{70}$ from the keypoint queries.

Decoder Heads. Three heads read the decoder output. The SimCC head h_{2D}, a two-layer MLP shared by all keypoints, turns each keypoint query into its two distributions, $\mathbf{P}_k = h_{\text{2D}}(\mathbf{z}_k)$, whose expectations over the bin centres give the 2D keypoint $\mathbf{x}_k$. Classifying the coordinates rather than regressing them keeps the image evidence of every joint in the decoder. The camera head h_{cam}, a linear layer, predicts the camera translation from the body query, $\mathbf{t} = h_{\text{cam}}(\mathbf{z}_0)$. The MHR head h_{MHR} predicts the body from the same query, $(\boldsymbol{\theta}, \boldsymbol{\beta}) = h_{\text{MHR}}(\mathbf{z}_0)$. It is a two-layer head in SAM 3D Body's continuous parameterisation: the root rotation as a 6D vector [47], the body pose in its continuous form, the identity coefficients $\boldsymbol{\beta}$, 28 bone-scale principal components and two hand blocks, which are decoded into the rig parameters $\boldsymbol{\theta}$. The root translation is fixed to zero (the person is placed by the camera translation), and the bone-scale coefficients and the six limb-length parameters are bounded to the range seen in training. Without these bounds, the scales enter the rig's forward kinematics multiplicatively along each limb, and a single large prediction can produce a skeleton hundreds of metres tall that diverges the training.

The 3D keypoints are not regressed: they are read from the posed rig through its own forward kinematics, so 3D keypoints and mesh can never disagree.

3.2. Training

Dataset. InstantHMR is trained on the human-annotated MHR fits that Meta released with SAM 3D Body [44], over the images of five of its sources: SA-1B [17], AI Challenger [42], Harmony4D [16], COCO [24] and MPII [1], 4.2 M person crops in total, sampled at 43 / 26 / 14 / 13 / 4 %. No 3DPW or EMDB images are used. The labels are the dataset's annotations, not the predictions of a teacher model, so the student is not capped at a teacher's accuracy.

Losses. The training loss is a weighted sum of eleven terms, which fall into three groups. The first group compares the predicted MHR parameters with the annotated ones: the joint parameters, the root rotation, the bone scales, the identity coefficients and the camera translation. The second group compares the geometry that the rig produces from these parameters: the 70 3D keypoints, 595 vertices spread over the mesh, and the 127 joints of the skeleton. As in NLF [34], the keypoint and vertex errors are plain Euclidean distances, not squared ones, so that a few bad samples do not dominate the gradient. The vertex term matters for shape: the identity coefficients $\boldsymbol{\beta}$ move the mesh but not the skeleton, so without it they receive no geometric gradient at all. The third group supervises the image side: the SimCC cross-entropy against a Gaussian label centred on each annotated keypoint, a distance on the 2D keypoints $\mathbf{x}$, and a reprojection term that projects the 3D keypoints with the predicted translation $\mathbf{t}$ and the calibrated camera, and compares them with the annotated 2D keypoints. The 40 finger keypoints weigh 0.2 in the 3D and reprojection terms, and the terms that go through the rig are ramped up over the first 2,000 steps. We tuned the weights of the eleven terms empirically, with single-batch overfitting tests and by balancing the measured gradient norm of each term; the equations and the weights are given in the supplementary material.

Method	Accuracy (mm) EMDB-1 PA-MPJPE	Latency per person (ms) RTX 4070 TRT fp16	RTX 4070 ORT fp32	RTX 4070 PyTorch	CPU i5 ORT	Phone NPU (fp16) S23	Phone NPU (fp16) S24	Phone CPU (fp32) S23	Phone CPU (fp32) A14
InstantHMR-M	48.8	**0.98**	4.42	–	**29.24**	**5.49**	**4.01**	**92.97**	**294**
Run on a phone NPU									
SMPLer-X-S32	71.0	1.38 1.4×	5.30	11.98	50.61	11.12 2×	8.16	164	509
MeshPose (HRNet-32)	73.7	2.36 2×	7.85	12.07	63.36	43.81 8×	31.19	309	979
NLF-S	45.0	1.52 1.6×	**3.71**	**10.19**	33.16	67.97 12×	54.45	103	306
HybrIK-R34	69.2	–	–	14.95	–	–	–	–	–
NLF-L	39.3	4.32 4×	10.62	16.40	136	125 23×	97.54	638	2,013
Multi-HMR-S	68.5	3.94 4×	24.66	32.12	387	139 25×	108	1,670	4,420
Do not run on a phone NPU									
CameraHMR	46.7	5.31 5×	13.87	21.61	408	×	×	–	×
TokenHMR	54.7	5.31 5×	14.56	23.36	409	×	×	–	×
HMR2.0b	78.4	5.51 6×	14.91	24.76	412	×	×	–	×
Fast SAM 3D Body	**38.6**	130 133×	–	192	–	×	×	–	×
SAM 3D Body	40.3	–	–	416	–	×	×	–	×

Table 1. **Latency on eight targets.** Model latency per person crop in ms, batch 1, up to a 6,890-vertex posed mesh. EMDB-1 PA-MPJPE (mm, 24 joints, Tab. 2) is shown for context. Methods are grouped by whether they run on a phone NPU at all; within a group, fastest first. Blue: ours. **Bold**: best in the column; underlined: best among the methods that run on a phone NPU. Shaded: real time on the phone ($\leq$ 33.3 ms, 30 FPS). Small grey factors: how many times slower than InstantHMR-M on the same target. ×: does not run on the device (memory over the limit, or the compiled model fails). –: the complete method could not be timed on this target; we report no partial timings (HybrIK's analytical IK does not export, so it is timed only in its own PyTorch release, IK included). Desktop: RTX 4070 with TensorRT fp16, ONNX Runtime (ORT) CUDA fp32 and each method's own PyTorch release; CPU: Intel i5-14400F, ORT on 6 performance cores. Phones: Samsung Galaxy S23 and S24 (Snapdragon 8 Gen 2 / Gen 3) NPU in fp16, and the S23 and Galaxy A14 CPUs in fp32, through Qualcomm AI Hub (median of 3 jobs). SAM 3D Body and Fast SAM 3D Body do not export as one graph: their PyTorch row is their own pipeline, and Fast's TensorRT row its own TensorRT backbone with the rest in PyTorch.

Training Recipe and Deployment. Each model is trained for 600 epochs of 400 k samples, batch 256 on four GPUs, with AdamW and a one-cycle learning-rate schedule peaking at 6×10^{-4}. Each training crop is either the standard crop (probability 0.75) or one of three other framings cut from the full image (zoomed out, zoomed in, shifted), with synthetic occlusion. This mix is the largest single gain we measured (Tab. 4): against training on the standard crop alone, it improves EMDB-1 PA-MPJPE by 10.0 mm (58.8 to 48.8) and 3DPW by 2.7 mm, more than an even or uniform mix of the framings. The 288 px input is the second: at every backbone size, it improves EMDB-1 PA-MPJPE by 4.4–5.5 mm over 224 px (Tab. 4 for the medium backbone, the supplementary material for the others).

The network exports to a single ONNX graph, which runs in fp16 on the S23 NPU with all of its 1,607 layers on the NPU. For the latency we report, the graph also includes the rig's forward kinematics and linear blend skinning up to the posed mesh $\mathbf{V}$.

4. Evaluation protocol

How a method is scored on 3DPW and EMDB moves its error by several millimetres, as much as the gaps between methods. The supplementary material shows this for InstantHMR-M alone: on EMDB-1, the choice of person boxes, focal length and body-model conversion moves its PA-MPJPE between 46.9 and 50.1 mm, and a learned conversion network [43] raises it to 65.9 mm. Picking the most favourable setting per benchmark would be protocol shopping. Our protocol is not the most favourable one for us: detector boxes score InstantHMR-M 1.9 mm better than the ground-truth boxes we report. We fix one protocol before scoring our models under it: the one of our closest competitors, the MHR-based SAM 3D Body and Fast SAM 3D Body, and score every method under it.

Settings. Every method is scored on the same frames, with the same person boxes and the same camera. The person boxes are the ones the benchmarks provide: EMDB ships one per frame, and for 3DPW, which ships none, we take the extent of the annotated 2D keypoints enlarged by 20 %. The only exception is NLF, which we run with its

Method	EMDB-1 (J24, mm)			3DPW (J14, mm)			RICH (J24, mm)			Latency (ms)	
	PA	MPJPE	PVE	PA	MPJPE	PVE	PA	MPJPE	PVE	RTX 4070	S23 NPU
InstantHMR-M	48.8	74.9	89.1	40.0	64.0	76.8	41.7	64.7	74.1	**0.98**	**5.49**
InstantHMR-M + det.	46.9	73.5	87.1	39.3	63.1	75.2	–	–	–	0.98	5.49
Run on a phone NPU											
SMPLer-X-S32†‡	71.0	109.8	127.7	50.4	78.4	91.9	61.2	99.5	112.2	1.38 1.4×	11.12 2×
MeshPose (HRNet-32)†	73.7	–	–	49.7	–	–	–	–	–	2.36 2×	43.81 8×
NLF-S‡	45.0	74.6	87.9	39.4	62.2	74.6	33.7	56.9	65.7	1.52 1.6×	67.97 12×
HybrIK-R34†	69.2	104.7	126.7	49.0	76.2	92.3	59.4	100.2	115.1	14.95^{p} 15×	–
NLF-L‡	39.3	67.7	79.7	35.1	57.2	68.3	**27.3**	**47.3**	**53.7**	4.32 4×	125 23×
Multi-HMR-S	68.5	103.0	115.6	54.8	84.4	99.1	69.4	102.4	112.5	3.94 4×	139 25×
Do not run on a phone NPU											
CameraHMR	46.7	76.7	91.5	38.8	61.3	72.6	33.6	54.8	63.8	5.31 5×	×
TokenHMR	54.7	88.8	106.5	44.5	69.8	83.5	42.1	80.6	93.6	5.31 5×	×
HMR2.0b	78.4	117.3	140.1	54.1	80.8	92.1	58.0	104.8	122.3	5.51 6×	×
Fast SAM 3D Body†	**38.6**	65.5	77.4	**34.1**	**55.0**	**66.3**	32.1	53.1	60.5	130^{t} 133×	×
SAM 3D Body†	40.3	**65.1**	**77.4**	35.1	55.7	67.2	33.5	54.6	62.3	416^{p} 425×	×

Table 2. **Accuracy under one protocol.** Every method is scored on the same frames with the same ground truth (Sec. 4): ground-truth person boxes, the calibrated focal length, and, for MHR methods, Meta's official MHR-to-SMPL conversion. Errors in mm on every 10th frame of EMDB-1 and 3DPW, and on the 21,287 test crops of RICH listed by CameraHMR, scored on the 24 joints regressed from the mesh. Groups and order as in Tab. 1. Blue: ours. **Bold**: best in the column; underlined: best among the methods that run on a phone NPU. Latency in ms; shaded : real time on the phone (≤ 33.3 ms, 30 FPS); small grey factors: how many times slower than InstantHMR-M. **InstantHMR-M + det.** is the same model given the boxes of an off-the-shelf person detector (YOLO11n-pose) instead of the ground-truth boxes. MeshPose outputs vertices without a metric scale, so only PA-MPJPE is defined. For the three methods that output only a mesh (SMPLer-X, Multi-HMR, MeshPose), the 24 EMDB-1 joints are regressed from the vertices; the other methods use the SMPL skeleton, as the ground truth does. †: trained on 3DPW-train; ‡: trained on RICH. ^{p}: its PyTorch release; ^{t}: its TensorRT backbone + PyTorch.

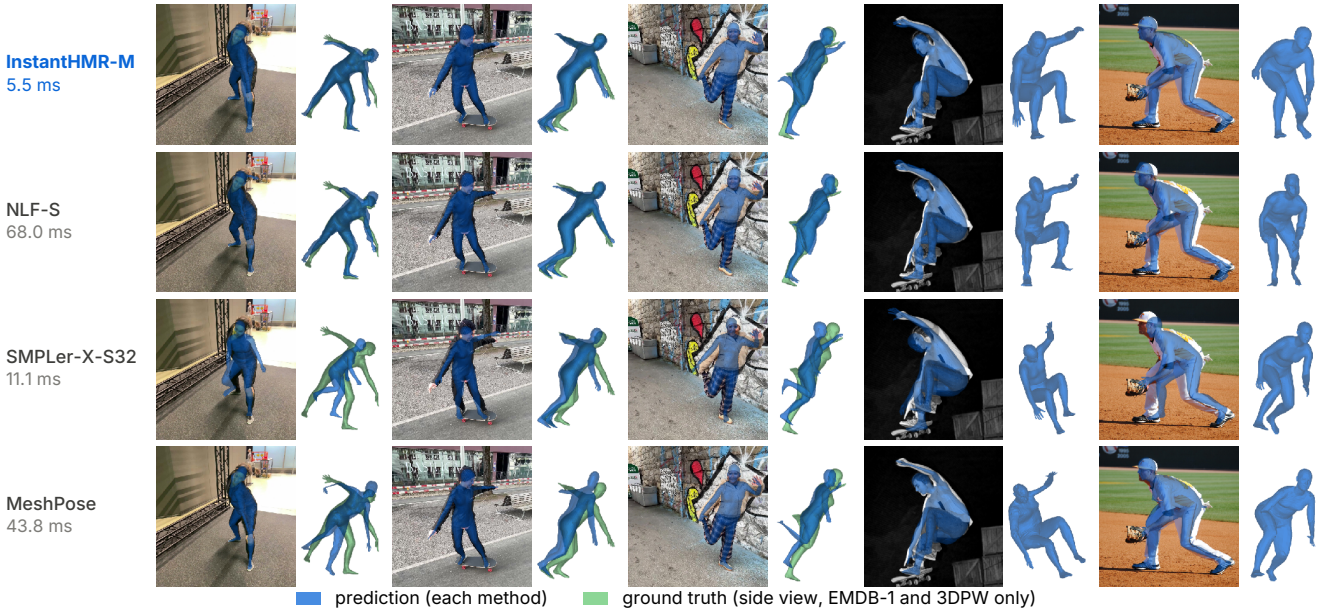

Figure 3. **Qualitative comparison of the methods that run on a phone NPU.** For each image, left: the predicted mesh projected into the image; right: the same mesh seen from the side. On the EMDB-1 [15] images (columns 1–3), the side view also shows the ground-truth mesh in green; the prediction is placed on its hips and scaled to its size, but not rotated. The COCO [24] images (columns 4–5) are in the wild and have no 3D ground truth. Under each method: its latency per person on the Samsung Galaxy S23 NPU.

	PA-MPJPE (mm)			Latency (ms)		
				RTX	S23	
Model	EMDB-1	3DPW	RICH	4070	NPU	CPU
InstantHMR-N	61.7	46.6	53.5	0.70	4.60	31.5
InstantHMR-T	58.0	44.0	49.7	0.73	4.84	37.4
InstantHMR-S	53.3	41.7	45.4	0.84	5.38	54.5
InstantHMR-M	48.8	40.0	41.7	0.98	5.49	93.0
InstantHMR-M-C	72.2	50.7	–	0.98	5.49	93.0

Table 3. **The InstantHMR family.** Nano, tiny, small and medium use HGNetV2-B0, B1, B2 and B4 backbones; *M-C* is the medium architecture trained only on commercially usable images (Harmony4D and COCO). Protocol of Tab. 2. Latency per person: RTX 4070 with TensorRT fp16, Samsung Galaxy S23 NPU in fp16 and CPU in fp32. Shaded : real time on the phone (≤ 33.3 ms).

own person detector, as in its paper; giving it the ground-truth boxes instead changes its error by less than 1 mm. The focal length is the calibrated one of each benchmark. Methods that output SMPL are scored as they are, and the others are first converted to SMPL: MHR outputs with Meta's official MHR-to-SMPL fitting, as in the Fast SAM 3D Body paper, and SMPL-X outputs through Meta's surface maps from SMPL-X to MHR to SMPL, which add 0.67 mm of error on average. As in each benchmark's literature, EMDB-1 is scored on the 24 SMPL joints and 3DPW on the 14 joints regressed from the mesh with the Human3.6M regressor. We score every 10th frame, which gives 2,419 frames on EMDB-1 and 3,562 person-frames on 3DPW; scoring every frame instead changes PA-MPJPE by at most 0.3 mm. On RICH [11], we score the 21,287 test crops listed in the label file of CameraHMR [29] (every 5th frame of the 50 test sequences, from all 7 cameras), with the person box that file gives. The ground truth is each crop's SMPL-X fit converted to SMPL and each model is scored on the 24 joints regressed from the SMPL mesh. A frame on which a method returns no prediction counts as missing.

Reproduction. We run each competitor from its released code and weights, with its own crop and preprocessing, and compare our score with the one printed in its paper, on the joint set of that paper. We call a method *reproduced* when the two PA-MPJPE differ by less than 1 mm. NLF-S and TokenHMR reproduce on both benchmarks, and CameraHMR on 3DPW. NLF-L scores 1.9–2.2 mm better than its paper, and SAM 3D Body, Fast SAM 3D Body, HybrIK, and CameraHMR on EMDB-1 score 1.3–4.5 mm worse. On RICH, CameraHMR reproduces its published number (0.4 mm better), SAM 3D Body scores 2.6 mm worse and Fast SAM 3D Body 1.7 mm better than their papers. HMR2.0b scores 9.9 mm worse than the run of it reported by SAM 3D Body, whose crop is not stated, while our harness reproduces HMR2.0b on EMDB-1 within 0.4 mm. Such gaps can come from settings that a paper does not fully specify, and we do not claim that any published number is wrong.

Latency. Latency is the model's time per person crop at batch 1, from the crop to a posed mesh of up to 6,890 vertices; detection and cropping are excluded for every method. Desktop timings are medians over interleaved rounds, so background load affects all methods alike. Phone timings use Qualcomm AI Hub [31] on physical devices, the median of three jobs. Only complete methods are timed: a method that does not export whole is timed in its own release (HybrIK, SAM 3D Body, Fast SAM 3D Body), and partial timings are left to the supplementary material.

5. Experiments

We evaluate on EMDB-1 [15], 3DPW test [40] and RICH test [11] under the protocol of Sec. 4. None of the three benchmarks' images are in InstantHMR's training data; four competitors trained on 3DPW-train (marked †), and three on RICH (‡), which makes EMDB-1 the fairer of the three.

Speed (Tab. 1). On the Galaxy S23 NPU, InstantHMR-M takes 5.49 ms per person, and 4.01 ms on the S24. Only one other method is real time on the phone: SMPLer-X-S32, at 11.12 ms, but 22 mm less accurate on EMDB-1. Every ViT-H and DINOv3-based method (CameraHMR, TokenHMR, HMR2.0, SAM 3D Body, Fast SAM 3D Body) fails to run on either phone NPU, for lack of memory or because the compiled model fails. On the desktop GPU, InstantHMR-M takes 0.98 ms with TensorRT, the fastest of all methods (nano: 0.70 ms).

Why InstantHMR is fast on the NPU. On the S23's CPU, InstantHMR-M (93 ms) and NLF-S (103 ms) are close, and on the desktop GPU they are 1.6× apart, but on the NPU NLF-S is 12× slower. Both models run entirely on the NPU (807 of 807 layers for NLF-S, 1,607 of 1,607 for InstantHMR-M), so the gap does not come from layers falling back to the CPU: it comes from how each model produces the mesh. NLF estimates each vertex as a separate query point with its own $8 \times 8 \times 8$ heatmap, whose score-weighted average gives the vertex position. For 6,890 vertices, most of the work is moving and summing tensors of $6{,}890 \times 512$ values, which an NPU, built for convolutions and matrix products, does slowly: gathers, reductions, transposes, reshapes and softmaxes take 71% of NLF-S's NPU time, its convolutions under 2%. With 1,048 query

points instead of 6,890, the same network runs in 8.1 ms. InstantHMR instead regresses the 204 MHR parameters once per person, with a convolutional backbone and a small decoder, operators the NPU is designed for; the mesh then follows from the rig's forward kinematics and skinning, a fixed weighted sum whose cost does not depend on the network.

Accuracy (Tab. 2). Among the methods that run on a phone, InstantHMR-M is 20–25 mm more accurate than SMPLer-X, MeshPose, HybrIK and Multi-HMR on EMDB-1 (PA-MPJPE), and 9–15 mm on 3DPW. MeshPose, the closest phone-oriented work, is released only with its HRNet-32 backbone, which scores 73.7 mm on EMDB-1 and 49.7 mm on 3DPW against 48.8 and 40.0 mm for InstantHMR-M; its paper reports its phone-sized variants, which are not released, 2–3 mm behind that model on 3DPW[2]. On hard poses, the gap can be much larger than these averages (Fig. 3). Only NLF is more accurate, at 12× (NLF-S) and 23× (NLF-L) the phone latency; on 3DPW, InstantHMR-M is within 0.6 mm of NLF-S. Without the phone constraint, Fast SAM 3D Body and SAM 3D Body remain the most accurate, 8–10 mm better than InstantHMR-M on EMDB-1, at 130–416 ms per person on a desktop GPU. With detector boxes in place of the ground-truth ones (InstantHMR-M + det.), InstantHMR-M improves to 46.9 mm on EMDB-1 and 39.3 mm on 3DPW, the setting NLF is evaluated in. RICH shows the same picture: InstantHMR-M (41.7 mm) is 18–28 mm more accurate than HybrIK, Multi-HMR and SMPLer-X, which is trained on RICH. Among the methods that run on a phone, only NLF, also trained on RICH, is more accurate; and CameraHMR, SAM 3D Body and Fast SAM 3D Body are 8–10 mm better than InstantHMR-M.

The family (Tab. 3). From nano to medium, EMDB-1 PA-MPJPE improves from 61.7 to 48.8 mm (53.5 to 41.7 mm on RICH), while the S23 NPU latency only grows from 4.60 to 5.49 ms: on the phone NPU, the latency barely depends on the backbone size, so medium is the natural choice. On the S23 CPU, it is a different story, the latency triples from nano to medium (31.5 to 93.0 ms), and nano is the only size that runs in real time there. On phones without a usable NPU, the smaller sizes are therefore worth the accuracy they lose. Training on commercially usable images only (Harmony4D and COCO, M-C) costs 23 mm on EMDB-1, with 0.5 M of the 4.2 M training crops: the training images matter more than any architectural change we tried.

[2]MeshPose-S / -XS, as published [18]: 3DPW PA-MPJPE 48.97 / 49.71 mm on 14 LSP joints from the authors' unreleased landmarker; 99 / 153 FPS on an iPhone 12 with CoreML. Weights not released; not comparable with our rows.

Training run	EMDB-1 PA	3DPW PA
Crop framing (B4, 288 px)		
standard crop only	58.8	42.7
standard 50 %	50.4	40.8
standard 75 % (InstantHMR-M)	48.8	40.0
uniform, 25 % each	51.2	41.1
Input size (B4, earlier recipe)		
224 px	54.2	43.9
288 px	49.8	42.2

Table 4. **Design ablation.** PA-MPJPE (mm) as in Tab. 2; each row is a full training run. Crop framing: the share of standard crops, the rest zoomed out, zoomed in or shifted. The input-size pair uses an earlier recipe (hence 49.8, not 48.8 mm); other backbones in the supplementary material.

6. Limitations and conclusion

Limitations. InstantHMR has no dedicated hand encoder or decoder. The hands are read from the same 288×288 crop of the whole person and the same 9×9 image tokens as the rest of the body, where a hand covers only a few pixels. SAM 3D Body, in contrast, refines the hands with a second decoder that runs on crops around each hand. Adding a light hand branch, without losing the real-time latency on the phone NPU, is left for future work. All our phone measurements are on Android phones with Qualcomm NPUs (Galaxy S23 and S24). We did not evaluate InstantHMR on Apple silicon (the Neural Engine of iPhones, through Core ML), which is also left for future work. InstantHMR estimates each frame on its own, and we report no temporal metric (jitter or acceleration error), although the applications we target are sensitive to it; evaluating it, with and without the 1 Euro filter of our released demo, is future work. Finally, InstantHMR is trained on the released annotations of the SAM 3D Body dataset, but many of its images, such as those of SA-1B and AI Challenger, may only be used for research. For this reason, the nano, tiny, small and medium models are released for research use only. InstantHMR-M-C, trained only on commercially usable images (Harmony4D and COCO), can be used in commercial products, but it is much less accurate: 72.2 mm PA-MPJPE on EMDB-1 against 48.8 mm for InstantHMR-M. Closing this gap requires more training images with permissive licences, real or synthetic.

Conclusion. We presented InstantHMR, a family of single-image MHR regressors built for phone NPUs. On the Samsung Galaxy S23 NPU, InstantHMR-M recovers a 6,890-vertex body mesh in 5.49 ms per person, at 48.8 mm PA-MPJPE on EMDB-1, 40.0 mm on 3DPW and 41.7 mm on RICH. It is the most accurate method that runs in real time on the phone; the two methods that are more accurate

and also run on a phone NPU, NLF-S and NLF-L, are 12× and 23× slower there. To make these comparisons fair, we re-evaluated eleven methods under one protocol, matched to the closest MHR-based competitors, and reported how closely each reproduces its paper. Code and models are available at https://github.com/Naturalpad-studio/InstantHMR.

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LLM Usage Statement

We used Claude Code (Anthropic), an LLM-based coding assistant, to write and run evaluation code under our direction (competitor reproduction, scoring, latency measurement, and the generation of tables and figures), to search the literature, and to draft and edit text. The model, its training and the experimental design are ours; we reviewed all code, verified every source and revised all text, and every number comes from saved result files.

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